

HealthCare Data Management System
A Project Report submitted
for
Database Management Systems (UCS310)

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Submitted to

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Introduction

The Healthcare Data Management System is a database-driven solution designed to efficiently manage patient records, treatment information, medical staff details, billing data, and disease statistics. With the increasing volume of healthcare data, maintaining records manually or using file-based systems leads to redundancy, inconsistency, and security issues.

A Database Management System (DBMS) ensures centralized data storage, fast retrieval, secure access, and reliable reporting. Compared to traditional file-based systems, a DBMS provides better concurrency control, data integrity through constraints, reduced redundancy, and backup and recovery mechanisms. The proposed system emphasizes backend implementation using SQL and PL/SQL to automate healthcare data processing and reporting.

Problem Statement

Traditional healthcare record management systems rely heavily on manual documentation or isolated file-based storage. These approaches suffer from:

- Data redundancy and inconsistency
- Difficulty in retrieving patient history
- Poor data security
- Lack of real-time statistical reporting
- No concurrency control for multiple users

The Healthcare Data Management System aims to develop a centralized, structured, and normalized database system that efficiently manages patient information, treatment records, billing details, and disease surveillance data.

Objectives of the Project

- To design an ER model for the Healthcare Data Management System
- To convert the ER model into relational schema
- To normalize the database up to Third Normal Form (3NF)
- To implement SQL queries for efficient data handling
- To use PL/SQL procedures, functions, triggers, and cursors
- To maintain data integrity using constraints and transactions

Scope of the Project

The project focuses on backend database implementation. It supports:

- Hospital Administrators
- Doctors
- Medical Staff

Major modules include:

- Patient Registration
- Doctor Management
- Appointment Scheduling
- Treatment Records
- Billing and Payments
- Disease Statistics Reporting

The system emphasizes database design and backend programming rather than graphical interface development.

Proposed System Description

The proposed system stores all healthcare-related data in a structured database. When a patient registers, their data is securely stored. Doctors can update diagnosis and treatment records. Billing information is automatically calculated using stored procedures. Triggers maintain data consistency and update statistics automatically.

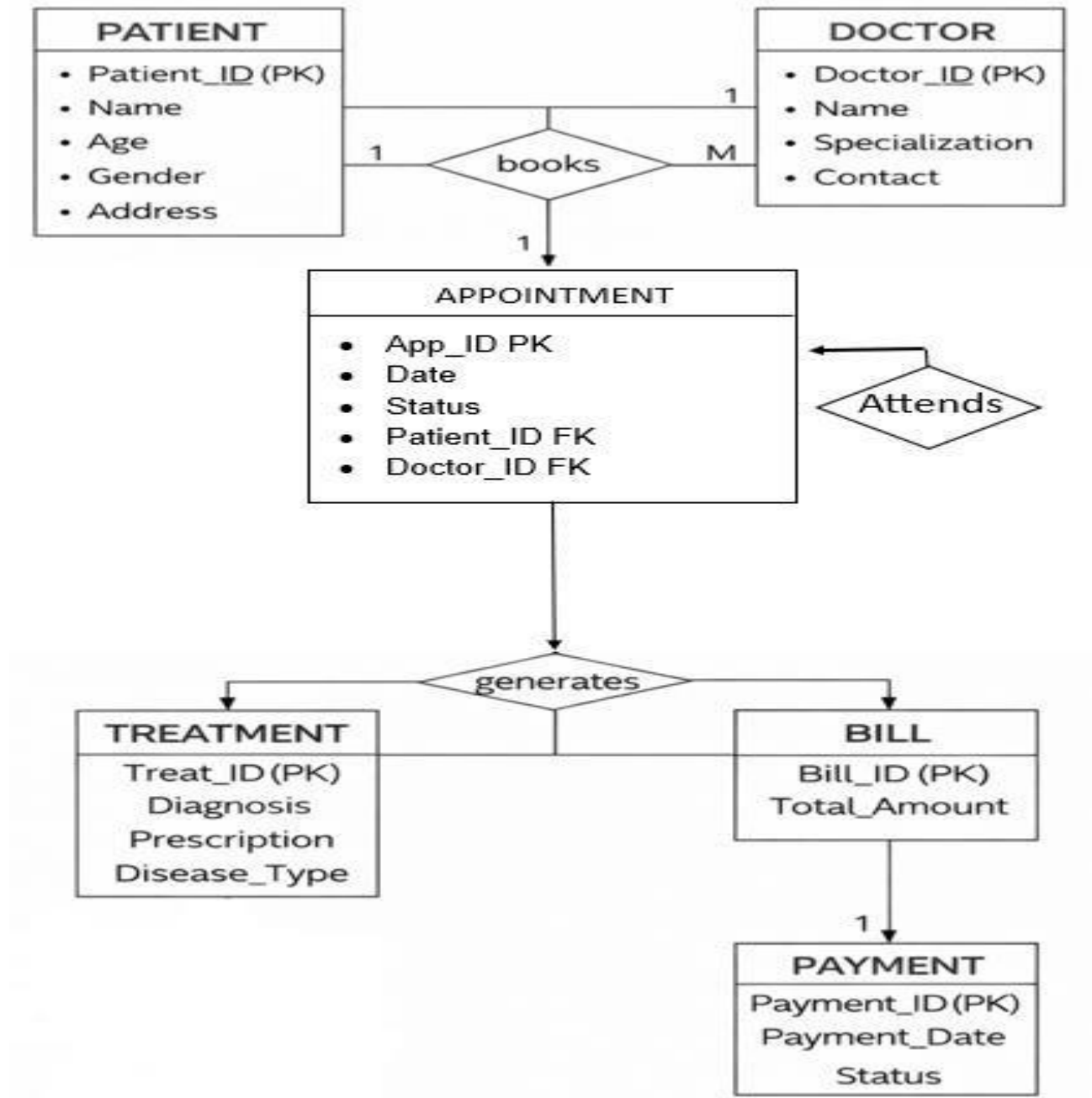
The system improves efficiency, reduces human error, ensures data integrity, and supports multi-user access with proper transaction control.

Database Design

The Healthcare Data Management System database is designed to efficiently manage patient information, doctor details, appointments, treatment records, billing, and disease statistics. The system is structured using an Entity–Relationship (ER) model that defines entities, attributes, relationships, cardinality, and constraints.

Primary keys and foreign keys are used to maintain referential integrity and ensure accurate data relationships. The database design reduces redundancy, improves consistency, and supports efficient retrieval of healthcare data

Entity–Relationship (ER) Diagram



Relationships and Cardinality

- A Patient books Appointment → (1 : M)
- A Doctor attends Appointment → (1 : M)
- An Appointment generates Treatment → (1 : 1)
- A Treatment generates Bill → (1 : 1)
- A Bill involves Payment → (1 : 1)

Relational Schema

The ER model is converted into relational tables as follows:

1. **PATIENT**

PATIENT(Patient_ID, Name, Age, Gender, Contact, Address)

2. **DOCTOR**

DOCTOR(Doctor_ID, Name, Specialization, Contact)

3. **APPOINTMENT**

APPOINTMENT(App_ID, Patient_ID, Doctor_ID, Date, Status)

FK: Patient_ID → PATIENT(Patient_ID)

FK: Doctor_ID → DOCTOR(Doctor_ID)

4. **TREATMENT**

TREATMENT(Treat_ID, App_ID, Diagnosis, Prescription, Disease_Type)

FK: App_ID → APPOINTMENT(App_ID)

5. **BILL**

BILL(Bill_ID, Treat_ID, Total_Amount)

FK: Treat_ID → TREATMENT(Treat_ID)

6. **PAYMENT**

PAYMENT(Payment_ID, Bill_ID, Payment_Date, Mode, Status)

FK: Bill_ID → BILL(Bill_ID)

Normalization

Normalization organizes database tables to reduce redundancy and improve data integrity. In the unnormalized form, healthcare data such as patient details, doctor information, appointment records, treatment data, and billing information may exist in a single large table, causing repetition and inconsistency.

First Normal Form (1NF) ensures atomic values and removes repeating groups. Each field contains single values only.

Second Normal Form (2NF) separates data into PATIENT, DOCTOR, APPOINTMENT, TREATMENT, BILL, and PAYMENT tables to eliminate partial dependencies.

Third Normal Form (3NF) removes transitive dependencies so that all non-key attributes depend only on their respective primary keys.

This results in an efficient, consistent, and well-structured healthcare database design.

Database Implementation

1. SQL Implementation

The Healthcare Data Management System database is implemented using SQL Data Definition Language (DDL) and Data Manipulation Language (DML).

DDL commands such as CREATE, ALTER, and DROP are used to design database tables and define relationships between entities. Primary keys and foreign keys are applied to maintain referential integrity.

DML commands such as INSERT, UPDATE, DELETE, and SELECT are used to store and retrieve healthcare data efficiently.

SELECT queries include:

- Joins between PATIENT, DOCTOR, and APPOINTMENT tables
- Subqueries for retrieving specific medical records
- Aggregate functions for generating treatment and billing statistics
- GROUP BY and HAVING clauses for reporting
- Views for simplified data access

Constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE, and CHECK are used to ensure data consistency and accuracy.

2. PL/SQL Components

The Healthcare Data Management System uses various PL/SQL components to automate operations and improve reliability.

Stored Procedures

Used for tasks such as patient registration, appointment scheduling, and bill generation.

Functions

Used to calculate total billing amount and generate medical statistics.

Triggers

Automatically execute when records are inserted or updated. For example, a trigger can update billing status after treatment entry.

Cursors

Used to retrieve and process multiple patient or treatment records for reporting purposes.

Exception Handling

Implemented to handle runtime errors and maintain database stability.

Transaction Management & Concurrency

Transaction management is used in the Healthcare Data Management System to ensure reliable and consistent database operations. SQL transaction control commands such as COMMIT, ROLLBACK, and SAVEPOINT are used to manage changes made during patient registration, appointment booking, treatment updates, and billing processes.

A COMMIT permanently saves successful transactions, while a ROLLBACK reverses incomplete or incorrect operations. SAVEPOINT allows partial rollback within a transaction when required.

Basic concurrency control mechanisms are applied to allow multiple users (such as doctors and administrators) to access the database simultaneously without conflicts. Locking techniques help prevent data inconsistency and ensure synchronized access to shared records.

Tools and Technologies Used

- **Frontend**: HTML, CSS, JavaScript, Bootstrap
- **Backend**: Express.js, Node.js
- **Database**: MySQL, Oracle, SQL, PL/SQL
- **Development Tools**: VS Code, MySQL Workbench, Oracle SQL Developer, Draw.io

Expected Outcomes

The Healthcare Data Management System results in a normalized and efficient database capable of managing patient records, doctor information, appointments, treatments, and billing data accurately.

The system enables faster data retrieval, automated billing processes, and improved medical reporting. It ensures data consistency and integrity through proper use of constraints and transactions.

Overall, the system improves operational efficiency, reduces redundancy, and supports secure multi-user access in a healthcare environment.

Conclusion

The Healthcare Data Management System applies DBMS concepts to efficiently manage patient, appointment, treatment, and billing data. The project demonstrates proper database design, normalization, and backend implementation using SQL and PL/SQL. It ensures data consistency, reduces redundancy, and supports reliable multi-user access in a healthcare environment.